Classwork Set A

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Problem 2

(b) Let us fix the vector $v_3 = (r, 0)$. We need to show that $v_i + v_j + v_3 = 0$ if and only if the angle between v_i and v_j is $120° = 2\pi/3$, and the angle between v_3 and v_i is also $120° = 2\pi/3$.

Proof.

$$\begin{aligned} & v_i + v_j + v_3 = 0 \\ \iff\ & v_i + v_j = -v_3 = \begin{bmatrix} -r \\ 0 \end{bmatrix} \end{aligned}$$

Let $\theta \in [-\pi, \pi]$ be the angle between v_3 and v_i, and let $\varphi \in [-\pi, \pi]$ be the angle between v_3 and v_j. The radius of the clock is r. We can find the components of the two vectors: $v_i = (r\cos\theta, r\sin\theta)$ and $v_j = (r\cos\varphi, r\sin\varphi)$.

$$\begin{aligned} \iff\ & \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix} + \begin{bmatrix} r\cos\varphi \\ r\sin\varphi \end{bmatrix} = \begin{bmatrix} -r \\ 0 \end{bmatrix} \\ \iff\ & \begin{cases} r\cos\theta + r\cos\varphi = -r \\ r\sin\theta + r\sin\varphi = 0 \end{cases} \end{aligned}$$

Notice: $r\sin\theta + r\sin\varphi = 0 \iff \sin\theta = -\sin\varphi$. Since $\theta, \varphi \in [-\pi, \pi]$, we have $\varphi = -\theta$.

$$\begin{aligned} \iff\ & r\cos\theta + r\cos(-\theta) = -r \\ \iff\ & 2\cos\theta = -1 \\ \iff\ & \cos\theta = -\frac{1}{2} \\ \iff\ & \theta = \pm\frac{2\pi}{3} \end{aligned}$$

Therefore, the angle between v_3 and v_i is $\theta = \pm 2\pi/3$, which can be either $2\pi/3$ or $-2\pi/3$. Either way, $\varphi = -\theta = \mp 2\pi/3$. As such, **the angle between v_i and v_3 is $120°$, as is the angle between v_j and v_3.** The angle between v_i and v_j is $2\pi - 2\pi/3 - 2\pi/3 = 2\pi/3 = 120°$. □

Problem 3

(a) True. If θ is the direction of u, v and m is the magnitude then $u = v = (m\cos\theta, m\sin\theta)$.

(b) False. If a is negative, then $\sqrt{2}a$ is negative but magnitude $\|(a, b)\|$ must be positive.

(c) True. u and v point in the same direction. u can be defined as $u = v/\|v\|$, and therefore:

$$\|v\|u = \cancel{\|v\|} \cdot \frac{v}{\cancel{\|v\|}} = v$$

(d) True. Since $\hat{i} = (1, 0)$ and $\hat{j} = (0, 1)$ are the standard unit vectors:

$$\|u\| = \|a\hat{i} + b\hat{j}\| = \|(a, 0) + (0, b)\| = \|(a, b)\| = \sqrt{a^2 + b^2} = 1 \implies a^2 + b^2 = 1$$

(e) True. If $v = a\hat{i} + b\hat{j} = 0$, then $v = (0, 0) \iff a = 0, b = 0 \implies a = -b$.

Problem 4

9. Let $c = -1$ and $\mathbf{v} = (1, 0)$. $\|c\mathbf{v}\| = \|(-1, 0)\| = 1 \neq -1 = (-1)\|(1, 0)\| = c\|\mathbf{v}\|$.